CALCULATION SUMMARY

Project Name : Injection Molding Plant

Project Location: Koregaon Bhima Pune

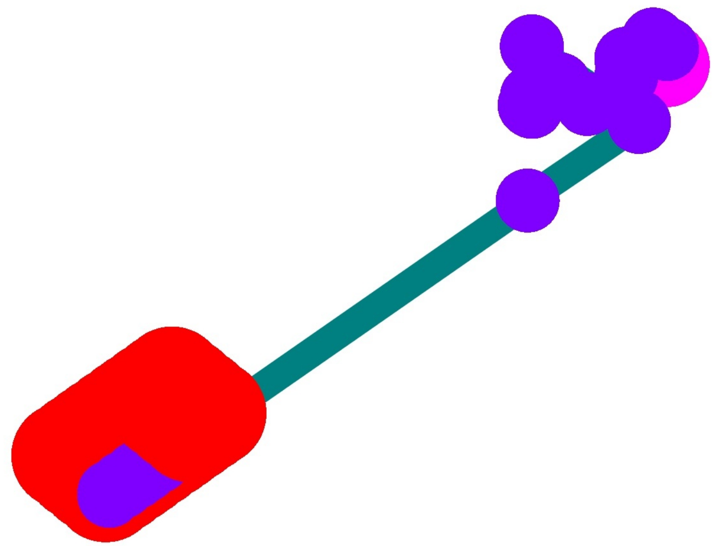
Contract No. :

City: pune

Design Areas

Design Area Name	Calc. Mode (Model)	Occupancy	Area of Application	Total Water	Pressure @ Source	Min. Density	Min. Pressure	Min. Flow	Calculated Heads	Hose Streams	Margin To Source
			(m²)	(l/min)	(bar)	(l/min/m²)	(bar)	(l/min)	#	(l/min)	(bar)
DesignArea_1	Demand (HW)		371.6	2225.19	Required 3.51	5	0.16	46.51	40	0	-3.51

Diagram for Initial System



Node Data

Node# Type	Hgroup Fitting	K-Fact. Stat. Pres.	Elev Coverage	X Y
		lpm/bar½ bar	m m²	m m
src1 Supply	SUPPLY		0	0 0
n2 Node	NODE us.90		3	0 0
n3 Node	NODE us.90		3	0 2.687
n4 Node	NODE us.90		3	-5.29 2.687
n5 Node	NODE us.90		-0.5	-5.29 2.687
n6 Node	NODE us.90		-0.5	-12.925 2.687
n7 Node	NODE us.90		-0.5	-12.925 8.16
n8 Node	NODE us.90		-0.5	-18.746 8.16
n9 Node	NODE us.90		-0.5	-18.746 7.714
n10 Node	NODE us.90		11.365	-18.746 7.714
n11 Node	NODE		0.5	-18.746 7.714
n12 Node	NODE		1.5	-18.746 7.714
n13 Node	NODE us.90		11.365	-18.746 -13.249
n22 Node	NODE us.90		12.048	-101.384 -13.249
n98 Node	NODE us.Tee		12.122	-101.384 -12.541
n99 Node	NODE us.Tee		12.405	-101.384 -9.856
n100 Node	NODE us.Tee		12.687	-101.384 -7.17
n40 Node	NODE us.90		12.969	-101.384 -4.485
n41 Node	NODE us.Tee		12.122	-122.384 -12.541
n42 Node	NODE us.Tee		12.405	-122.384 -9.856
n43 Node	NODE us.Tee		12.687	-122.384 -7.17
n44 Node	NODE us.90		12.969	-122.384 -4.485
n45 Node	NODE us.Tee		12.122	-119.784 -12.541
n46 Node	NODE us.Tee		12.405	-119.784 -9.856
n47 Node	NODE us.Tee		12.687	-119.784 -7.17

Node Data

Node# Type	Hgroup Fitting	K-Fact. Stat. Pres.	Elev Coverage	X Y
		lpm/bar½ bar	m m²	m m
n48 Node	NODE us.90		12.969	-119.784 -4.485
n49 Node	NODE us.Tee		12.122	-117.184 -12.541
n50 Node	NODE us.Tee		12.405	-117.184 -9.856
n51 Node	NODE us.Tee		12.687	-117.184 -7.17
n52 Node	NODE us.90		12.969	-117.184 -4.485
n53 Node	NODE us.Tee		12.122	-115.384 -12.541
n54 Node	NODE us.Tee		12.405	-115.384 -9.856
n55 Node	NODE us.Tee		12.687	-115.384 -7.17
n56 Node	NODE us.90		12.969	-115.384 -4.485
n57 Node	NODE us.Tee		12.122	-112.784 -12.541
n58 Node	NODE us.Tee		12.405	-112.784 -9.856
n59 Node	NODE us.Tee		12.687	-112.784 -7.17
n60 Node	NODE us.90		12.969	-112.784 -4.485
n87 Node	NODE us.90		12.048	-110.184 -13.249
n61 Node	NODE us.Tee		12.122	-110.184 -12.541
n62 Node	NODE us.Tee		12.405	-110.184 -9.856
n63 Node	NODE us.Tee		12.687	-110.184 -7.17
n64 Node	NODE us.90		12.969	-110.184 -4.485
n36 Node	NODE us.90		12.048	-108.384 -13.249
n65 Node	NODE us.Tee		12.122	-108.384 -12.541
n66 Node	NODE us.Tee		12.405	-108.384 -9.856
n67 Node	NODE us.Tee		12.687	-108.384 -7.17
n68 Node	NODE us.90		12.969	-108.384 -4.485
h1 Overhead Sprinkler	HEAD	115.3	13.069 9.3	-122.384 -4.485
h2 Overhead Sprinkler	HEAD	115.3	12.787 9.3	-122.384 -7.17

Node Data

Node# Type	Hgroup Fitting	K-Fact. Stat. Pres.	Elev Coverage	X Y
		lpm/bar½ bar	m m²	m m
h3 Overhead Sprinkler	HEAD	115.3	12.505 9.3	-122.384 -9.856
h4 Overhead Sprinkler	HEAD	115.3	12.222 9.3	-122.384 -12.541
h5 Overhead Sprinkler	HEAD	115.3	12.222 9.3	-119.784 -12.541
h6 Overhead Sprinkler	HEAD	115.3	12.505 9.3	-119.784 -9.856
h7 Overhead Sprinkler	HEAD	115.3	12.787 9.3	-119.784 -7.17
h8 Overhead Sprinkler	HEAD	115.3	13.069 9.3	-119.784 -4.485
h9 Overhead Sprinkler	HEAD	115.3	13.069 9.3	-117.184 -4.485
h10 Overhead Sprinkler	HEAD	115.3	12.787 9.3	-117.184 -7.17
h11 Overhead Sprinkler	HEAD	115.3	12.505 9.3	-117.184 -9.856
h12 Overhead Sprinkler	HEAD	115.3	12.222 9.3	-117.184 -12.541
h13 Overhead Sprinkler	HEAD	115.3	12.222 9.3	-115.384 -12.541
h14 Overhead Sprinkler	HEAD	115.3	12.505 9.3	-115.384 -9.856
h15 Overhead Sprinkler	HEAD	115.3	12.787 9.3	-115.384 -7.17
h16 Overhead Sprinkler	HEAD	115.3	13.069 9.3	-115.384 -4.485
h17 Overhead Sprinkler	HEAD	115.3	13.069 9.3	-112.784 -4.485
h18 Overhead Sprinkler	HEAD	115.3	12.787 9.3	-112.784 -7.17
h19 Overhead Sprinkler	HEAD	115.3	12.505 9.3	-112.784 -9.856
h20 Overhead Sprinkler	HEAD	115.3	12.222 9.3	-112.784 -12.541
h21 Overhead Sprinkler	HEAD	115.3	12.222 9.3	-110.184 -12.541
h22 Overhead Sprinkler	HEAD	115.3	12.505 9.3	-110.184 -9.856
h23 Overhead Sprinkler	HEAD	115.3	12.787 9.3	-110.184 -7.17
h24 Overhead Sprinkler	HEAD	115.3	13.069 9.3	-110.184 -4.485
h25 Overhead Sprinkler	HEAD	115.3	13.069 9.3	-108.384 -4.485
h26 Overhead Sprinkler	HEAD	115.3	12.787 9.3	-108.384 -7.17
h27 Overhead Sprinkler	HEAD	115.3	12.505 9.3	-108.384 -9.856

Node Data

Node# Type	Hgroup Fitting	K-Fact. Stat. Pres.	Elev Coverage	X Y
		lpm/bar½ bar	m m²	m m
h28 Overhead Sprinkler	HEAD	115.3	12.222 9.3	-108.384 -12.541
h37 Overhead Sprinkler	HEAD	115.3	12.222 9.3	-101.384 -12.541
h38 Overhead Sprinkler	HEAD	115.3	12.505 9.3	-101.384 -9.856
h39 Overhead Sprinkler	HEAD	115.3	12.787 9.3	-101.384 -7.17
h40 Overhead Sprinkler	HEAD	115.3	13.069 9.3	-101.384 -4.485
n69 Node	NODE		11.365	-40.534 -13.249
n71 Node	NODE us.Tee		11.365	-108.384 -13.249
n86 Node	NODE us.Tee		11.365	-110.184 -13.249
n76 Node	NODE us.90		11.365	-122.384 -13.249
n26 Node	NODE us.90		12.048	-122.384 -13.249
n75 Node	NODE us.Tee		11.365	-119.784 -13.249
n28 Node	NODE us.90		12.048	-119.784 -13.249
n74 Node	NODE us.Tee		11.365	-117.184 -13.249
n30 Node	NODE us.90		12.048	-117.184 -13.249
n73 Node	NODE us.Tee		11.365	-115.384 -13.249
n32 Node	NODE us.90		12.048	-115.384 -13.249
n72 Node	NODE us.Tee		11.365	-112.784 -13.249
n34 Node	NODE us.90		12.048	-112.784 -13.249
n96 Node	NODE us.Tee		11.365	-101.384 -13.249
n88 Node	NODE us.Tee		11.365	-105.784 -13.249
n91 Node	NODE us.90		12.048	-105.784 -13.249
n37 Node	NODE us.Tee		12.122	-105.784 -12.541
n38 Node	NODE us.Tee		12.405	-105.784 -9.856
n39 Node	NODE us.Tee		12.687	-105.784 -7.17
n95 Node	NODE us.90		12.969	-105.784 -4.485

Node Data

Node# Type	Hgroup Fitting	K-Fact. Stat. Pres.	Elev Coverage	X Y
		lpm/bar½ bar	m m²	m m
h29 Overhead Sprinkler	HEAD	115.3	12.222 9.3	-105.784 -12.541
h30 Overhead Sprinkler	HEAD	115.3	12.505 9.3	-105.784 -9.856
h31 Overhead Sprinkler	HEAD	115.3	12.787 9.3	-105.784 -7.17
h32 Overhead Sprinkler	HEAD	115.3	13.069 9.3	-105.784 -4.485
n89 Node	NODE us.Tee		11.365	-103.184 -13.249
n97 Node	NODE us.90		12.048	-103.184 -13.249
n92 Node	NODE us.Tee		12.122	-103.184 -12.541
n93 Node	NODE us.Tee		12.405	-103.184 -9.856
n94 Node	NODE us.Tee		12.687	-103.184 -7.17
n101 Node	NODE us.90		12.969	-103.184 -4.485
h33 Overhead Sprinkler	HEAD	115.3	12.222 9.3	-103.184 -12.541
h34 Overhead Sprinkler	HEAD	115.3	12.505 9.3	-103.184 -9.856
h35 Overhead Sprinkler	HEAD	115.3	12.787 9.3	-103.184 -7.17
h36 Overhead Sprinkler	HEAD	115.3	13.069 9.3	-103.184 -4.485

Pipe Data

Pipe Ref.	Type	Start	End	Size	HWC	Length	Fittings	Eq.Len.	Total Len.	Schedule
						m		m	m	
p1	Pipe	src1	n2	6	120	3				40
p2	Pipe	n2	n3	6	120	2.687				40
p3	Pipe	n3	n4	6	120	5.29				40
p4	Pipe	n4	n5	6	120	3.5				40
p5	Pipe	n5	n6	6	120	7.635				40
p6	Pipe	n6	n7	6	120	5.473				40
p7	Pipe	n7	n8	6	120	5.821				40
p8	Pipe	n8	n9	6	120	0.446				40
p10	Pipe	n9	n11	6	120	1				40
v1	Valve	n11	n12	6	0	1				AV-1 Check
p13	Pipe	n12	n10	6	120	9.865				40
p14	Pipe	n10	n13	6	120	20.963				40
b87	Brline	n96	n22	1.5	120	0.683				40
b88	Brline	n22	n98	1.5	120	0.712				40
b89	Brline	n98	n99	1.5	120	2.7				40
b91	Brline	n99	n100	1.5	120	2.7				40
b93	Brline	n100	n40	1.5	120	2.7				40
b77	Brline	n86	n87	1.5	120	0.683				40
b2	Brline	n26	n41	1.5	120	0.712				40
b3	Brline	n41	n42	1.5	120	2.7				40
b4	Brline	n42	n43	1.5	120	2.7				40
b5	Brline	n43	n44	1.5	120	2.7				40
b7	Brline	n28	n45	1.5	120	0.712				40
b8	Brline	n45	n46	1.5	120	2.7				40
b9	Brline	n46	n47	1.5	120	2.7				40
b10	Brline	n47	n48	1.5	120	2.7				40
b12	Brline	n30	n49	1.5	120	0.712				40
b13	Brline	n49	n50	1.5	120	2.7				40
b14	Brline	n50	n51	1.5	120	2.7				40
b15	Brline	n51	n52	1.5	120	2.7				40
b17	Brline	n32	n53	1.5	120	0.712				40
b18	Brline	n53	n54	1.5	120	2.7				40
b19	Brline	n54	n55	1.5	120	2.7				40
b20	Brline	n55	n56	1.5	120	2.7				40
b22	Brline	n34	n57	1.5	120	0.712				40
b23	Brline	n57	n58	1.5	120	2.7				40
b24	Brline	n58	n59	1.5	120	2.7				40
b25	Brline	n59	n60	1.5	120	2.7				40
b27	Brline	n87	n61	1.5	120	0.712				40
b28	Brline	n61	n62	1.5	120	2.7				40
b29	Brline	n62	n63	1.5	120	2.7				40
b30	Brline	n63	n64	1.5	120	2.7				40
b31	Brline	n71	n36	1.5	120	0.683				40
b32	Brline	n36	n65	1.5	120	0.712				40

Pipe Data

Pipe Ref.	Type	Start	End	Size	HWC	Length	Fittings	Eq.Len.	Total Len.	Schedule
						m		m	m	
b33	Brline	n65	n66	1.5	120	2.7				40
b34	Brline	n66	n67	1.5	120	2.7				40
b35	Brline	n67	n68	1.5	120	2.7				40
b41	Brline	n44	h1	1	120	0.1				40
b42	Brline	n43	h2	1	120	0.1				40
b43	Brline	n42	h3	1	120	0.1				40
b44	Brline	n41	h4	1	120	0.1				40
b45	Brline	n45	h5	1	120	0.1				40
b46	Brline	n46	h6	1	120	0.1				40
b47	Brline	n47	h7	1	120	0.1				40
b48	Brline	n48	h8	1	120	0.1				40
b49	Brline	n52	h9	1	120	0.1				40
b50	Brline	n51	h10	1	120	0.1				40
b51	Brline	n50	h11	1	120	0.1				40
b52	Brline	n49	h12	1	120	0.1				40
b53	Brline	n53	h13	1	120	0.1				40
b54	Brline	n54	h14	1	120	0.1				40
b55	Brline	n55	h15	1	120	0.1				40
b56	Brline	n56	h16	1	120	0.1				40
b57	Brline	n60	h17	1	120	0.1				40
b58	Brline	n59	h18	1	120	0.1				40
b59	Brline	n58	h19	1	120	0.1				40
b60	Brline	n57	h20	1	120	0.1				40
b61	Brline	n61	h21	1	120	0.1				40
b62	Brline	n62	h22	1	120	0.1				40
b63	Brline	n63	h23	1	120	0.1				40
b64	Brline	n64	h24	1	120	0.1				40
b65	Brline	n68	h25	1	120	0.1				40
b66	Brline	n67	h26	1	120	0.1				40
b67	Brline	n66	h27	1	120	0.1				40
b68	Brline	n65	h28	1	120	0.1				40
b90	Brline	n98	h37	1	120	0.1				40
b92	Brline	n99	h38	1	120	0.1				40
b94	Brline	n100	h39	1	120	0.1				40
b95	Brline	n40	h40	1	120	0.1				40
p23	Pipe	n13	n69	6	120	21.788				40
p25	Pipe	n88	n71	4	120	2.6				40
p26	Pipe	n71	n86	4	120	1.8				40
p27	Pipe	n86	n72	4	120	2.6				40
p28	Pipe	n72	n73	4	120	2.6				40
p29	Pipe	n73	n74	4	120	1.8				40
p30	Pipe	n74	n75	4	120	2.6				40
p31	Pipe	n75	n76	4	120	2.6				40
b1	Brline	n76	n26	1.5	120	0.683				40

Pipe Data

Pipe Ref.	Type	Start	End	Size	HWC	Length	Fittings	Eq.Len.	Total Len.	Schedule
						m		m	m	
b73	Brline	n75	n28	1.5	120	0.683				40
b74	Brline	n74	n30	1.5	120	0.683				40
b75	Brline	n73	n32	1.5	120	0.683				40
b76	Brline	n72	n34	1.5	120	0.683				40
p33	Pipe	n89	n88	4	120	2.6				40
p34	Pipe	n69	n96	4	120	60.85				40
p35	Pipe	n96	n89	4	120	1.8				40
b36	Brline	n88	n91	1.5	120	0.683				40
b37	Brline	n91	n37	1.5	120	0.712				40
b38	Brline	n37	n38	1.5	120	2.7				40
b69	Brline	n37	h29	1	120	0.1				40
b39	Brline	n38	n39	1.5	120	2.7				40
b70	Brline	n38	h30	1	120	0.1				40
b40	Brline	n39	n95	1.5	120	2.7				40
b71	Brline	n39	h31	1	120	0.1				40
b72	Brline	n95	h32	1	120	0.1				40
b78	Brline	n89	n97	1.5	120	0.683				40
b79	Brline	n97	n92	1.5	120	0.712				40
b80	Brline	n92	n93	1.5	120	2.7				40
b81	Brline	n92	h33	1	120	0.1				40
b82	Brline	n93	n94	1.5	120	2.7				40
b83	Brline	n93	h34	1	120	0.1				40
b84	Brline	n94	n101	1.5	120	2.7				40
b85	Brline	n94	h35	1	120	0.1				40
b86	Brline	n101	h36	1	120	0.1				40

HYDRAULIC CALCULATIONS
for

Job Information

Project Name : Injection Molding Plant	
Contract No. :	City: pune
Project Location: Koregaon Bhima Pune	Date: 31-12-2025

Contractor Information

Name of Contractor:	
Address:	City:
Phone Number:	E-mail:
Name of Designer:	
Authority Having Jurisdiction:	

Design

Remote Area Name	DesignArea_1
Remote Area Location	
Occupancy Classification	
Density (l/min/m²)	5
Area of Application (m²)	371.6
Coverage per Sprinkler (m²)	9.3
Number of Calculated Sprinklers	40
In-Rack Demand (l/min)	0
Special Heads	
Hose Streams (l/min)	0
Total Water Required (incl. Hose Streams) (l/min)	2225.19
Required Pressure at Source (bar)	3.51
Type of System	Wet
Volume - Entire System (l)	2429.5 l

Water Supply Information

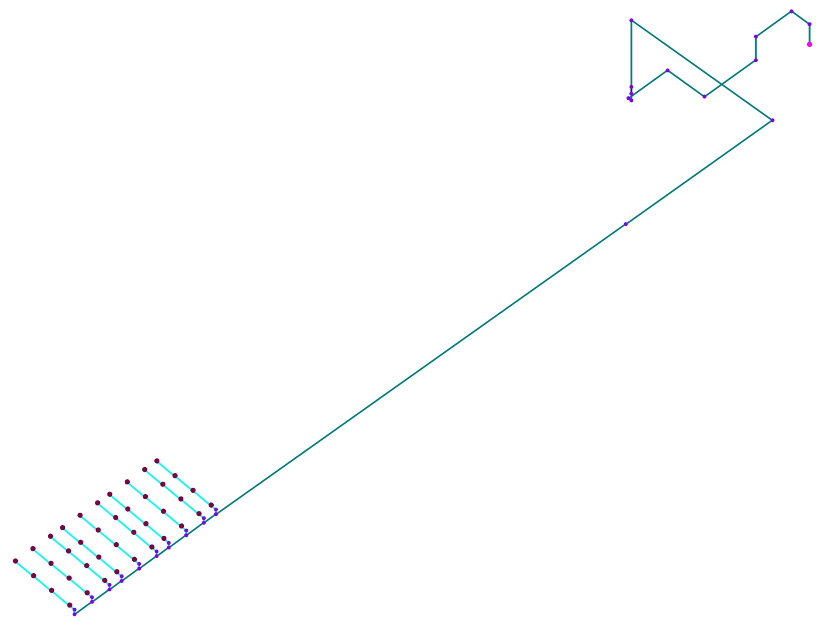
Date

Location

Source

Notes

Diagram for Design Area : DesignArea_1



Hydraulic Analysis for : DesignArea_1

Calculation Info

Calculation Mode	Demand
Hydraulic Model	Hazen-Williams
Fluid Name	Water @ 60F (15.6C)
Fluid Weight, (N/m³)	N/A for Hazen-Williams calculation.
Fluid Dynamic Viscosity, (Pa·s)	N/A for Hazen-Williams calculation.

Water Supply Parameters

Hoses

Inside Hose Flow / Standpipe Demand (l/min)	
Outside Hose Flow (l/min)	
Additional Outside Hose Flow (l/min)	
Other (custom defined) Hose Flow (l/min)	
Total Hose Flow (l/min)	

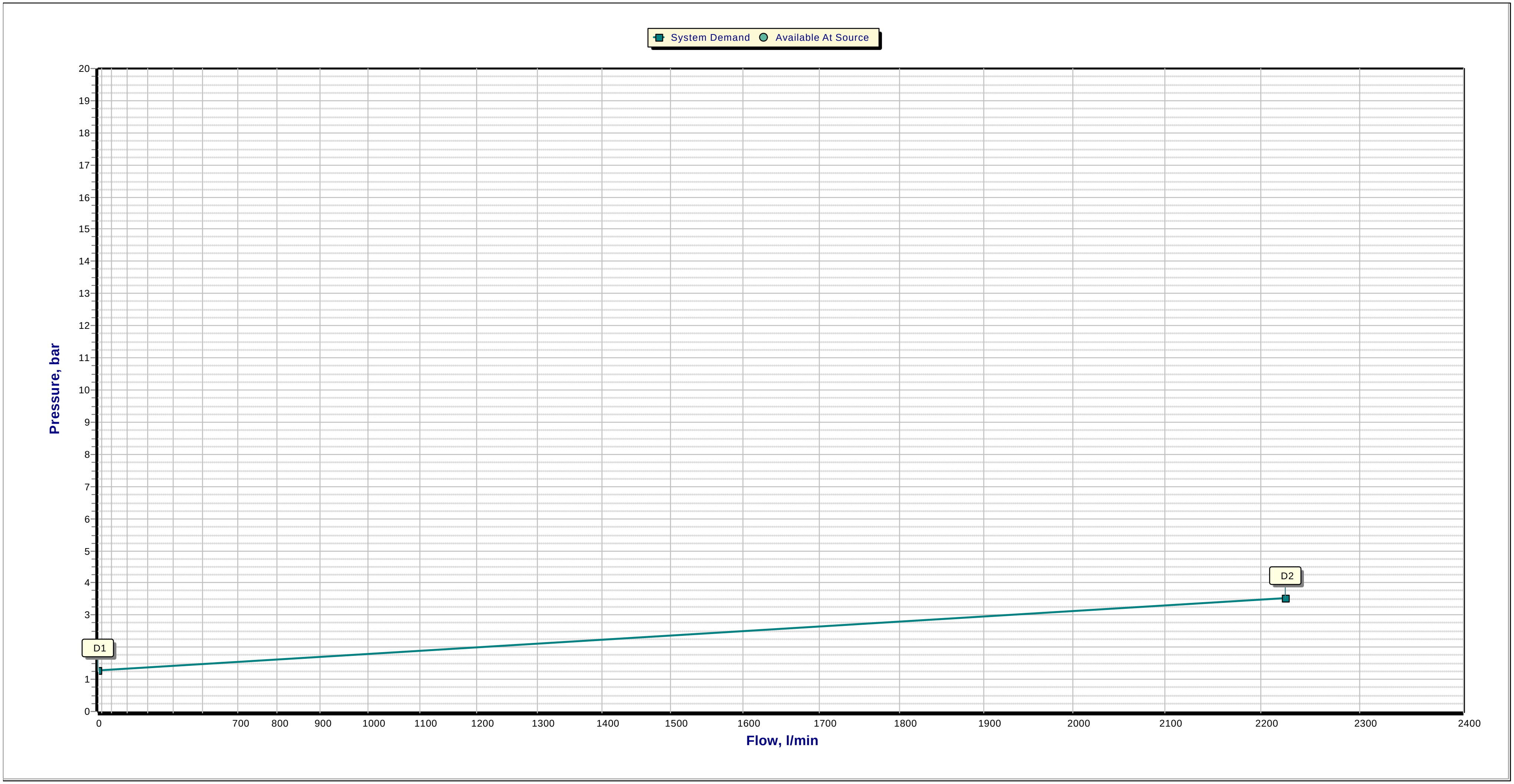
Sprinklers

Ovehead Sprinkler Flow (l/min)	2225.19
InRack Sprinkler Flow (l/min)	0
Other (custom defined) Sprinkler Flow (l/min)	0
Total Sprinkler Flow (l/min)	2225.19

Other

Required Margin of Safety (bar)	0
src1 - Pressure (bar)	3.51
src1 - Flow (l/min)	2225.19
Demand w/o System Pump(s)	N/A

Hydraulic Analysis for : DesignArea_1



Hydraulic Analysis for : DesignArea_1

Graph Labels

Label	Description	Values	
		Flow (l/min)	Pressure (bar)
D1	Elevation Pressure	0	1.28
D2	System Demand	2225.19	3.51

Open Heads

Head Ref.	Head Type	Coverage	K-Factor	Required			Calculated		
				Density	Flow	Pressure	Density	Flow	Pressure
		(m²)	(lpm/bar½)	(l/min/m²)	(l/min)	(bar)	(l/min/m²)	(l/min)	(bar)
h1	Overhead Sprinkler	9.3	115.3	5	46.5	0.16	5.3	48.79	0.18
h2	Overhead Sprinkler	9.3	115.3	5	46.5	0.16	5.5	51.2	0.2
h3	Overhead Sprinkler	9.3	115.3	5	46.5	0.16	6	56.14	0.24
h4	Overhead Sprinkler	9.3	115.3	5	46.5	0.16	6.8	62.74	0.3
h5	Overhead Sprinkler	9.3	115.3	5	46.5	0.16	6.5	60.63	0.28
h6	Overhead Sprinkler	9.3	115.3	5	46.5	0.16	5.8	54.09	0.22
h7	Overhead Sprinkler	9.3	115.3	5	46.5	0.16	5.3	49.11	0.18
h8	Overhead Sprinkler	9.3	115.3	5	46.5	0.16	5	46.51	0.16
h9	Overhead Sprinkler	9.3	115.3	5	46.5	0.16	5	46.71	0.16
h10	Overhead Sprinkler	9.3	115.3	5	46.5	0.16	5.3	49.3	0.18
h11	Overhead Sprinkler	9.3	115.3	5	46.5	0.16	5.8	54.27	0.22
h12	Overhead Sprinkler	9.3	115.3	5	46.5	0.16	6.5	60.82	0.28
h13	Overhead Sprinkler	9.3	115.3	5	46.5	0.16	6.6	61.09	0.28
h14	Overhead Sprinkler	9.3	115.3	5	46.5	0.16	5.9	54.54	0.22
h15	Overhead Sprinkler	9.3	115.3	5	46.5	0.16	5.3	49.57	0.18
h16	Overhead Sprinkler	9.3	115.3	5	46.5	0.16	5.1	47.01	0.17
h17	Overhead Sprinkler	9.3	115.3	5	46.5	0.16	5.1	47.72	0.17

h18	Overhead Sprinkler	9.3	115.3	5	46.5	0.16	5.4	50.22	0.19
h19	Overhead Sprinkler	9.3	115.3	5	46.5	0.16	5.9	55.18	0.23
h20	Overhead Sprinkler	9.3	115.3	5	46.5	0.16	6.6	61.75	0.29
h21	Overhead Sprinkler	9.3	115.3	5	46.5	0.16	6.8	62.74	0.3
h22	Overhead Sprinkler	9.3	115.3	5	46.5	0.16	6	56.14	0.24
h23	Overhead Sprinkler	9.3	115.3	5	46.5	0.16	5.5	51.21	0.2
h24	Overhead Sprinkler	9.3	115.3	5	46.5	0.16	5.3	48.79	0.18
h25	Overhead Sprinkler	9.3	115.3	5	46.5	0.16	5.4	49.81	0.19
h26	Overhead Sprinkler	9.3	115.3	5	46.5	0.16	5.6	52.15	0.2
h27	Overhead Sprinkler	9.3	115.3	5	46.5	0.16	6.1	57.07	0.24
h28	Overhead Sprinkler	9.3	115.3	5	46.5	0.16	6.9	63.69	0.31
h37	Overhead Sprinkler	9.3	115.3	5	46.5	0.16	7.5	69.76	0.37
h38	Overhead Sprinkler	9.3	115.3	5	46.5	0.16	6.8	62.94	0.3
h39	Overhead Sprinkler	9.3	115.3	5	46.5	0.16	6.3	58.09	0.25
h40	Overhead Sprinkler	9.3	115.3	5	46.5	0.16	6.1	56.21	0.24
h29	Overhead Sprinkler	9.3	115.3	5	46.5	0.16	7.1	65.51	0.32
h30	Overhead Sprinkler	9.3	115.3	5	46.5	0.16	6.3	58.83	0.26
h31	Overhead Sprinkler	9.3	115.3	5	46.5	0.16	5.8	53.94	0.22
h32	Overhead Sprinkler	9.3	115.3	5	46.5	0.16	5.6	51.75	0.2
h33	Overhead Sprinkler	9.3	115.3	5	46.5	0.16	7.3	67.8	0.35
h34	Overhead Sprinkler	9.3	115.3	5	46.5	0.16	6.6	61.05	0.28
h35	Overhead Sprinkler	9.3	115.3	5	46.5	0.16	6	56.18	0.24
h36	Overhead Sprinkler	9.3	115.3	5	46.5	0.16	5.8	54.17	0.22

Node Data

Node# Elev	Type Hgroup	K-Fact. Open/Closed	Discharge Overdischarge	Coverage Density	Tot. Pres. Elev. Pres.	Req. Pres. Req. Discharge
m		lpm/bar½	l/min l/min	m² l/min/m²	bar bar	bar l/min
src1 0	Supply SUPPLY		-2225.19		3.51 0	
n2 3	Node NODE				3.21 -0.29	
n3 3	Node NODE				3.19 -0.29	
n4 3	Node NODE				3.16 -0.29	
n5 -0.5	Node NODE				3.48 0.05	
n6 -0.5	Node NODE				3.45 0.05	
n7 -0.5	Node NODE				3.42 0.05	
n8 -0.5	Node NODE				3.39 0.05	
n9 -0.5	Node NODE				3.37 0.05	
n10 11.365	Node NODE				2.15 -1.11	
n11 0.5	Node NODE				3.26 -0.05	
n12 1.5	Node NODE				3.14 -0.15	
n13 11.365	Node NODE				2.07 -1.11	
n22 12.048	Node NODE				0.49 -1.18	
n98 12.122	Node NODE				0.42 -1.19	
n99 12.405	Node NODE				0.34 -1.22	
n100 12.687	Node NODE				0.29 -1.24	
n40 12.969	Node NODE				0.26 -1.27	
n41 12.122	Node NODE				0.34 -1.19	
n42 12.405	Node NODE				0.27 -1.22	
n43 12.687	Node NODE				0.23 -1.24	
n44 12.969	Node NODE				0.2 -1.27	
n45 12.122	Node NODE				0.32 -1.19	
n46 12.405	Node NODE				0.26 -1.22	
n47 12.687	Node NODE				0.21 -1.24	

Node Data

Node# Elev	Type Hgroup	K-Fact. Open/Closed	Discharge Overdischarge	Coverage Density	Tot. Pres. Elev. Pres.	Req. Pres. Req. Discharge
m		lpm/bar½	l/min l/min	m² l/min/m²	bar bar	bar l/min
n48 12.969	Node NODE				0.18 -1.27	
n49 12.122	Node NODE				0.32 -1.19	
n50 12.405	Node NODE				0.26 -1.22	
n51 12.687	Node NODE				0.21 -1.24	
n52 12.969	Node NODE				0.18 -1.27	
n53 12.122	Node NODE				0.32 -1.19	
n54 12.405	Node NODE				0.26 -1.22	
n55 12.687	Node NODE				0.22 -1.24	
n56 12.969	Node NODE				0.18 -1.27	
n57 12.122	Node NODE				0.33 -1.19	
n58 12.405	Node NODE				0.27 -1.22	
n59 12.687	Node NODE				0.22 -1.24	
n60 12.969	Node NODE				0.19 -1.27	
n87 12.048	Node NODE				0.4 -1.18	
n61 12.122	Node NODE				0.34 -1.19	
n62 12.405	Node NODE				0.27 -1.22	
n63 12.687	Node NODE				0.23 -1.24	
n64 12.969	Node NODE				0.2 -1.27	
n36 12.048	Node NODE				0.41 -1.18	
n65 12.122	Node NODE				0.35 -1.19	
n66 12.405	Node NODE				0.28 -1.22	
n67 12.687	Node NODE				0.24 -1.24	
n68 12.969	Node NODE				0.21 -1.27	
h1 13.069	Overhead Sprinkler HEAD	115.3 Open	48.79 2.29	9.3 5.3	0.18 -1.28	0.16 46.5
h2 12.787	Overhead Sprinkler HEAD	115.3 Open	51.2 4.7	9.3 5.5	0.2 -1.25	0.16 46.5

Node Data

Node# Elev	Type Hgroup	K-Fact. Open/Closed	Discharge Overdischarge	Coverage Density	Tot. Pres. Elev. Pres.	Req. Pres. Req. Discharge
m		lpm/bar½	l/min l/min	m² l/min/m²	bar bar	bar l/min
h3 12.505	Overhead Sprinkler HEAD	115.3 Open	56.14 9.64	9.3 6	0.24 -1.23	0.16 46.5
h4 12.222	Overhead Sprinkler HEAD	115.3 Open	62.74 16.24	9.3 6.8	0.3 -1.2	0.16 46.5
h5 12.222	Overhead Sprinkler HEAD	115.3 Open	60.63 14.13	9.3 6.5	0.28 -1.2	0.16 46.5
h6 12.505	Overhead Sprinkler HEAD	115.3 Open	54.09 7.59	9.3 5.8	0.22 -1.23	0.16 46.5
h7 12.787	Overhead Sprinkler HEAD	115.3 Open	49.11 2.61	9.3 5.3	0.18 -1.25	0.16 46.5
h8 13.069	Overhead Sprinkler HEAD	115.3 Open	46.51 0.01	9.3 5	0.16 -1.28	0.16 46.5
h9 13.069	Overhead Sprinkler HEAD	115.3 Open	46.71 0.21	9.3 5	0.16 -1.28	0.16 46.5
h10 12.787	Overhead Sprinkler HEAD	115.3 Open	49.3 2.8	9.3 5.3	0.18 -1.25	0.16 46.5
h11 12.505	Overhead Sprinkler HEAD	115.3 Open	54.27 7.77	9.3 5.8	0.22 -1.23	0.16 46.5
h12 12.222	Overhead Sprinkler HEAD	115.3 Open	60.82 14.32	9.3 6.5	0.28 -1.2	0.16 46.5
h13 12.222	Overhead Sprinkler HEAD	115.3 Open	61.09 14.59	9.3 6.6	0.28 -1.2	0.16 46.5
h14 12.505	Overhead Sprinkler HEAD	115.3 Open	54.54 8.04	9.3 5.9	0.22 -1.23	0.16 46.5
h15 12.787	Overhead Sprinkler HEAD	115.3 Open	49.57 3.07	9.3 5.3	0.18 -1.25	0.16 46.5
h16 13.069	Overhead Sprinkler HEAD	115.3 Open	47.01 0.51	9.3 5.1	0.17 -1.28	0.16 46.5
h17 13.069	Overhead Sprinkler HEAD	115.3 Open	47.72 1.22	9.3 5.1	0.17 -1.28	0.16 46.5
h18 12.787	Overhead Sprinkler HEAD	115.3 Open	50.22 3.72	9.3 5.4	0.19 -1.25	0.16 46.5
h19 12.505	Overhead Sprinkler HEAD	115.3 Open	55.18 8.68	9.3 5.9	0.23 -1.23	0.16 46.5
h20 12.222	Overhead Sprinkler HEAD	115.3 Open	61.75 15.25	9.3 6.6	0.29 -1.2	0.16 46.5
h21 12.222	Overhead Sprinkler HEAD	115.3 Open	62.74 16.24	9.3 6.8	0.3 -1.2	0.16 46.5
h22 12.505	Overhead Sprinkler HEAD	115.3 Open	56.14 9.64	9.3 6	0.24 -1.23	0.16 46.5
h23 12.787	Overhead Sprinkler HEAD	115.3 Open	51.21 4.71	9.3 5.5	0.2 -1.25	0.16 46.5
h24 13.069	Overhead Sprinkler HEAD	115.3 Open	48.79 2.29	9.3 5.3	0.18 -1.28	0.16 46.5
h25 13.069	Overhead Sprinkler HEAD	115.3 Open	49.81 3.31	9.3 5.4	0.19 -1.28	0.16 46.5
h26 12.787	Overhead Sprinkler HEAD	115.3 Open	52.15 5.65	9.3 5.6	0.2 -1.25	0.16 46.5
h27 12.505	Overhead Sprinkler HEAD	115.3 Open	57.07 10.57	9.3 6.1	0.24 -1.23	0.16 46.5

Node Data

Node# Elev	Type Hgroup	K-Fact. Open/Closed	Discharge Overdischarge	Coverage Density	Tot. Pres. Elev. Pres.	Req. Pres. Req. Discharge
m		lpm/bar½	l/min l/min	m² l/min/m²	bar bar	bar l/min
h28 12.222	Overhead Sprinkler HEAD	115.3 Open	63.69 17.19	9.3 6.9	0.31 -1.2	0.16 46.5
h37 12.222	Overhead Sprinkler HEAD	115.3 Open	69.76 23.26	9.3 7.5	0.37 -1.2	0.16 46.5
h38 12.505	Overhead Sprinkler HEAD	115.3 Open	62.94 16.44	9.3 6.8	0.3 -1.23	0.16 46.5
h39 12.787	Overhead Sprinkler HEAD	115.3 Open	58.09 11.59	9.3 6.3	0.25 -1.25	0.16 46.5
h40 13.069	Overhead Sprinkler HEAD	115.3 Open	56.21 9.71	9.3 6.1	0.24 -1.28	0.16 46.5
n69 11.365	Node NODE				1.99 -1.11	
n71 11.365	Node NODE				0.56 -1.11	
n86 11.365	Node NODE				0.55 -1.11	
n76 11.365	Node NODE				0.51 -1.11	
n26 12.048	Node NODE				0.4 -1.18	
n75 11.365	Node NODE				0.51 -1.11	
n28 12.048	Node NODE				0.37 -1.18	
n74 11.365	Node NODE				0.52 -1.11	
n30 12.048	Node NODE				0.37 -1.18	
n73 11.365	Node NODE				0.52 -1.11	
n32 12.048	Node NODE				0.38 -1.18	
n72 11.365	Node NODE				0.53 -1.11	
n34 12.048	Node NODE				0.38 -1.18	
n96 11.365	Node NODE				0.66 -1.11	
n88 11.365	Node NODE				0.59 -1.11	
n91 12.048	Node NODE				0.43 -1.18	
n37 12.122	Node NODE				0.37 -1.19	
n38 12.405	Node NODE				0.3 -1.22	
n39 12.687	Node NODE				0.25 -1.24	
n95 12.969	Node NODE				0.22 -1.27	

Node Data

Node# Elev	Type Hgroup	K-Fact. Open/Closed	Discharge Overdischarge	Coverage Density	Tot. Pres. Elev. Pres.	Req. Pres. Req. Discharge
m		lpm/bar½	l/min l/min	m² l/min/m²	bar bar	bar l/min
h29 12.222	Overhead Sprinkler HEAD	115.3 Open	65.51 19.01	9.3 7.1	0.32 -1.2	0.16 46.5
h30 12.505	Overhead Sprinkler HEAD	115.3 Open	58.83 12.33	9.3 6.3	0.26 -1.23	0.16 46.5
h31 12.787	Overhead Sprinkler HEAD	115.3 Open	53.94 7.44	9.3 5.8	0.22 -1.25	0.16 46.5
h32 13.069	Overhead Sprinkler HEAD	115.3 Open	51.75 5.25	9.3 5.6	0.2 -1.28	0.16 46.5
n89 11.365	Node NODE				0.62 -1.11	
n97 12.048	Node NODE				0.46 -1.18	
n92 12.122	Node NODE				0.39 -1.19	
n93 12.405	Node NODE				0.32 -1.22	
n94 12.687	Node NODE				0.27 -1.24	
n101 12.969	Node NODE				0.24 -1.27	
h33 12.222	Overhead Sprinkler HEAD	115.3 Open	67.8 21.3	9.3 7.3	0.35 -1.2	0.16 46.5
h34 12.505	Overhead Sprinkler HEAD	115.3 Open	61.05 14.55	9.3 6.6	0.28 -1.23	0.16 46.5
h35 12.787	Overhead Sprinkler HEAD	115.3 Open	56.18 9.68	9.3 6	0.24 -1.25	0.16 46.5
h36 13.069	Overhead Sprinkler HEAD	115.3 Open	54.17 7.67	9.3 5.8	0.22 -1.28	0.16 46.5

Pipe Data

Path # Pipe Ref.	Type Hgroup	Schedule Size	HWC Rough.	Fittings Eq.Len.	Length Total Len.	Flow Velocity	Fr.Resist. Loss Frict.	Vel.Pres. Loss Elev.	Start End	Start Disch. End Disch.	Start Tot.Pres. End Tot.Pres.
			mm	m	m m	l/min m/s	bar/m bar	bar bar		l/min l/min	bar bar
1 b48	Brline PIPE	40 1	120 0.1016	1(us.90); 0.61	0.1 0.71	46.51 1.39	0.0119 0.01	0.01 0.01	n48 h8	 46.51	0.18 0.16
1 b10	Brline PIPE	40 1.5	120 0.1016	1(us.Tee-Run);	2.7	46.51 0.59	0.0015 0.00	0.00 0.03	n47 n48		0.21 0.18
1 b9	Brline PIPE	40 1.5	120 0.1016	1(us.Tee-Run);	2.7	95.62 1.21	0.0056 0.02	0.01 0.03	n46 n47		0.26 0.21
1 b8	Brline PIPE	40 1.5	120 0.1016	1(us.Tee-Run);	2.7	149.71 1.9	0.0129 0.03	0.02 0.03	n45 n46		0.32 0.26
1 b7	Brline PIPE	40 1.5	120 0.1016	1(us.90); 1.219	0.712 1.931	210.34 2.67	0.0242 0.05	0.04 0.01	n28 n45		0.37 0.32
1 b73	Brline PIPE	40 1.5	120 0.1016	1(us.Tee-Br); 2.438	0.683 3.121	210.34 2.67	0.0242 0.08	0.04 0.07	n75 n28		0.51 0.37
1 p30	Pipe PIPE	40 4	120 0.1016	1(us.Tee-Run);	2.6	429.2 0.87	0.001 0.00	0.00 0	n74 n75		0.52 0.51
1 p29	Pipe PIPE	40 4	120 0.1016	1(us.Tee-Run);	1.8	640.3 1.3	0.0022 0.00	0.01 0	n73 n74		0.52 0.52
1 p28	Pipe PIPE	40 4	120 0.1016	1(us.Tee-Run);	2.6	852.5 1.73	0.0037 0.01	0.01 0	n72 n73		0.53 0.52
1 p27	Pipe PIPE	40 4	120 0.1016	1(us.Tee-Run);	2.6	1067.38 2.17	0.0056 0.01	0.02 0	n86 n72		0.55 0.53
1 p26	Pipe PIPE	40 4	120 0.1016	1(us.Tee-Run);	1.8	1286.26 2.61	0.008 0.01	0.03 0	n71 n86		0.56 0.55
1 p25	Pipe PIPE	40 4	120 0.1016	1(us.Tee-Run);	2.6	1508.98 3.06	0.0107 0.03	0.05 0	n88 n71		0.59 0.56
1 p33	Pipe PIPE	40 4	120 0.1016	1(us.Tee-Run);	2.6	1739 3.53	0.0139 0.04	0.06 0	n89 n88		0.62 0.59
1 p35	Pipe PIPE	40 4	120 0.1016	1(us.Tee-Run);	1.8	1978.2 4.01	0.0177 0.03	0.08 0	n96 n89		0.66 0.62
1 p34	Pipe PIPE	40 4	120 0.1016		60.85	2225.19 4.52	0.022 1.34	0.1 0	n69 n96		1.99 0.66
1 p23	Pipe PIPE	40 6	120 0.1016	1(us.90); 4.267	21.788 26.055	2225.19 1.99	0.003 0.08	0.02 0	n13 n69		2.07 1.99
1 p14	Pipe PIPE	40 6	120 0.1016	1(us.90); 4.267	20.963 25.23	2225.19 1.99	0.003 0.08	0.02 0	n10 n13		2.15 2.07
1 p13	Pipe PIPE	40 6	120 0.1016		9.865	2225.19 1.99	0.003 0.03	0.02 0.97	n12 n10		3.14 2.15
1 v1	Valve VALVE	AV-1 Check 6	0 0		1	2225.19 0	0.0197 0.02	0 0.1	n11 n12		3.26 3.14
1 p10	Pipe PIPE	40 6	120 0.1016	1(us.90); 4.267	1 5.267	2225.19 1.99	0.003 0.02	0.02 0.1	n9 n11		3.37 3.26
1 p8	Pipe PIPE	40 6	120 0.1016	1(us.90); 4.267	0.446 4.713	2225.19 1.99	0.003 0.01	0.02 0	n8 n9		3.39 3.37
1 p7	Pipe PIPE	40 6	120 0.1016	1(us.90); 4.267	5.821 10.088	2225.19 1.99	0.003 0.03	0.02 0	n7 n8		3.42 3.39
1 p6	Pipe PIPE	40 6	120 0.1016	1(us.90); 4.267	5.473 9.74	2225.19 1.99	0.003 0.03	0.02 0	n6 n7		3.45 3.42
1 p5	Pipe PIPE	40 6	120 0.1016	1(us.90); 4.267	7.635 11.902	2225.19 1.99	0.003 0.04	0.02 0	n5 n6		3.48 3.45
1 p4	Pipe PIPE	40 6	120 0.1016	1(us.90); 4.267	3.5 7.767	2225.19 1.99	0.003 0.02	0.02 -0.34	n4 n5		3.16 3.48

Pipe Data

Path # Pipe Ref.	Type Hgroup	Schedule Size	HWC Rough.	Fittings Eq.Len.	Length Total Len.	Flow Velocity	Fr.Resist. Loss Frict.	Vel.Pres. Loss Elev.	Start End	Start Disch. End Disch.	Start Tot.Pres. End Tot.Pres.
			mm	m	m m	l/min m/s	bar/m bar	bar bar		l/min l/min	bar bar
1 p3	Pipe PIPE	40 6	120 0.1016	1(us.90); 4.267	5.29 9.557	2225.19 1.99	0.003 0.03	0.02 0	n3 n4		3.19 3.16
1 p2	Pipe PIPE	40 6	120 0.1016	1(us.90); 4.267	2.687 6.954	2225.19 1.99	0.003 0.02	0.02 0	n2 n3		3.21 3.19
1 p1	Pipe PIPE	40 6	120 0.1016		3	2225.19 1.99	0.003 0.01	0.02 0.29	src1 n2	-2225.19	3.51 3.21
2 b49	Brline PIPE	40 1	120 0.1016	1(us.90); 0.61	0.1 0.71	46.71 1.4	0.012 0.01	0.01 0.01	n52 h9	 46.71	0.18 0.16
2 b15	Brline PIPE	40 1.5	120 0.1016	1(us.Tee-Run);	2.7	46.71 0.59	0.0015 0.00	0.00 0.03	n51 n52		0.21 0.18
2 b14	Brline PIPE	40 1.5	120 0.1016	1(us.Tee-Run);	2.7	96.01 1.22	0.0057 0.02	0.01 0.03	n50 n51		0.26 0.21
2 b13	Brline PIPE	40 1.5	120 0.1016	1(us.Tee-Run);	2.7	150.28 1.91	0.013 0.04	0.02 0.03	n49 n50		0.32 0.26
2 b12	Brline PIPE	40 1.5	120 0.1016	1(us.90); 1.219	0.712 1.931	211.1 2.68	0.0243 0.05	0.04 0.01	n30 n49		0.37 0.32
2 b74	Brline PIPE	40 1.5	120 0.1016	1(us.Tee-Br); 2.438	0.683 3.121	211.1 2.68	0.0243 0.08	0.04 0.07	n74 n30		0.52 0.37
3 b56	Brline PIPE	40 1	120 0.1016	1(us.90); 0.61	0.1 0.71	47.01 1.41	0.0121 0.01	0.01 0.01	n56 h16	 47.01	0.18 0.17
3 b20	Brline PIPE	40 1.5	120 0.1016	1(us.Tee-Run);	2.7	47.01 0.6	0.0015 0.00	0.00 0.03	n55 n56		0.22 0.18
3 b19	Brline PIPE	40 1.5	120 0.1016	1(us.Tee-Run);	2.7	96.57 1.23	0.0057 0.02	0.01 0.03	n54 n55		0.26 0.22
3 b18	Brline PIPE	40 1.5	120 0.1016	1(us.Tee-Run);	2.7	151.11 1.92	0.0131 0.04	0.02 0.03	n53 n54		0.32 0.26
3 b17	Brline PIPE	40 1.5	120 0.1016	1(us.90); 1.219	0.712 1.931	212.2 2.69	0.0246 0.05	0.04 0.01	n32 n53		0.38 0.32
3 b75	Brline PIPE	40 1.5	120 0.1016	1(us.Tee-Br); 2.438	0.683 3.121	212.2 2.69	0.0246 0.08	0.04 0.07	n73 n32		0.52 0.38
4 b57	Brline PIPE	40 1	120 0.1016	1(us.90); 0.61	0.1 0.71	47.72 1.43	0.0125 0.01	0.01 0.01	n60 h17	 47.72	0.19 0.17
4 b25	Brline PIPE	40 1.5	120 0.1016	1(us.Tee-Run);	2.7	47.72 0.61	0.0015 0.00	0.00 0.03	n59 n60		0.22 0.19
4 b24	Brline PIPE	40 1.5	120 0.1016	1(us.Tee-Run);	2.7	97.95 1.24	0.0059 0.02	0.01 0.03	n58 n59		0.27 0.22
4 b23	Brline PIPE	40 1.5	120 0.1016	1(us.Tee-Run);	2.7	153.13 1.94	0.0134 0.04	0.02 0.03	n57 n58		0.33 0.27
4 b22	Brline PIPE	40 1.5	120 0.1016	1(us.90); 1.219	0.712 1.931	214.88 2.73	0.0251 0.05	0.04 0.01	n34 n57		0.38 0.33
4 b76	Brline PIPE	40 1.5	120 0.1016	1(us.Tee-Br); 2.438	0.683 3.121	214.88 2.73	0.0251 0.08	0.04 0.07	n72 n34		0.53 0.38
5 b41	Brline PIPE	40 1	120 0.1016	1(us.90); 0.61	0.1 0.71	48.79 1.46	0.013 0.01	0.01 0.01	n44 h1	 48.79	0.2 0.18
5 b5	Brline PIPE	40 1.5	120 0.1016	1(us.Tee-Run);	2.7	48.79 0.62	0.0016 0.00	0.00 0.03	n43 n44		0.23 0.2
5 b4	Brline PIPE	40 1.5	120 0.1016	1(us.Tee-Run);	2.7	99.99 1.27	0.0061 0.02	0.01 0.03	n42 n43		0.27 0.23
5 b3	Brline PIPE	40 1.5	120 0.1016	1(us.Tee-Run);	2.7	156.13 1.98	0.0139 0.04	0.02 0.03	n41 n42		0.34 0.27

Pipe Data

Path # Pipe Ref.	Type Hgroup	Schedule Size	HWC Rough.	Fittings Eq.Len.	Length Total Len.	Flow Velocity	Fr.Resist. Loss Frict.	Vel.Pres. Loss Elev.	Start End	Start Disch. End Disch.	Start Tot.Pres. End Tot.Pres.
			mm	m	m m	l/min m/s	bar/m bar	bar bar		l/min l/min	bar bar
5 b2	Brline PIPE	40 1.5	120 0.1016	1(us.90); 1.219	0.712 1.931	218.87 2.78	0.026 0.05	0.04 0.01	n26 n41		0.4 0.34
5 b1	Brline PIPE	40 1.5	120 0.1016	1(us.90); 1.219	0.683 1.902	218.87 2.78	0.026 0.05	0.04 0.07	n76 n26		0.51 0.4
5 p31	Pipe PIPE	40 4	120 0.1016	1(us.Tee-Run);	2.6	218.87 0.44	0.0003 0	0 0	n75 n76		0.51 0.51
6 b64	Brline PIPE	40 1	120 0.1016	1(us.90); 0.61	0.1 0.71	48.79 1.46	0.013 0.01	0.01 0.01	n64 h24	 48.79	0.2 0.18
6 b30	Brline PIPE	40 1.5	120 0.1016	1(us.Tee-Run);	2.7	48.79 0.62	0.0016 0.00	0.00 0.03	n63 n64		0.23 0.2
6 b29	Brline PIPE	40 1.5	120 0.1016	1(us.Tee-Run);	2.7	100 1.27	0.0061 0.02	0.01 0.03	n62 n63		0.27 0.23
6 b28	Brline PIPE	40 1.5	120 0.1016	1(us.Tee-Run);	2.7	156.14 1.98	0.0139 0.04	0.02 0.03	n61 n62		0.34 0.27
6 b27	Brline PIPE	40 1.5	120 0.1016	1(us.90); 1.219	0.712 1.931	218.88 2.78	0.026 0.05	0.04 0.01	n87 n61		0.4 0.34
6 b77	Brline PIPE	40 1.5	120 0.1016	1(us.Tee-Br); 2.438	0.683 3.121	218.88 2.78	0.026 0.08	0.04 0.07	n86 n87		0.55 0.4
7 b47	Brline PIPE	40 1	120 0.1016	1(us.Tee-Br); 1.524	0.1 1.624	49.11 1.47	0.0132 0.02	0.01 0.01	n47 h7	 49.11	0.21 0.18
8 b50	Brline PIPE	40 1	120 0.1016	1(us.Tee-Br); 1.524	0.1 1.624	49.3 1.47	0.0133 0.02	0.01 0.01	n51 h10	 49.3	0.21 0.18
9 b55	Brline PIPE	40 1	120 0.1016	1(us.Tee-Br); 1.524	0.1 1.624	49.57 1.48	0.0134 0.02	0.01 0.01	n55 h15	 49.57	0.22 0.18
10 b65	Brline PIPE	40 1	120 0.1016	1(us.90); 0.61	0.1 0.71	49.81 1.49	0.0135 0.01	0.01 0.01	n68 h25	 49.81	0.21 0.19
10 b35	Brline PIPE	40 1.5	120 0.1016	1(us.Tee-Run);	2.7	49.81 0.63	0.0017 0.00	0.00 0.03	n67 n68		0.24 0.21
10 b34	Brline PIPE	40 1.5	120 0.1016	1(us.Tee-Run);	2.7	101.96 1.29	0.0063 0.02	0.01 0.03	n66 n67		0.28 0.24
10 b33	Brline PIPE	40 1.5	120 0.1016	1(us.Tee-Run);	2.7	159.03 2.02	0.0144 0.04	0.02 0.03	n65 n66		0.35 0.28
10 b32	Brline PIPE	40 1.5	120 0.1016	1(us.90); 1.219	0.712 1.931	222.72 2.83	0.0269 0.05	0.04 0.01	n36 n65		0.41 0.35
10 b31	Brline PIPE	40 1.5	120 0.1016	1(us.Tee-Br); 2.438	0.683 3.121	222.72 2.83	0.0269 0.08	0.04 0.07	n71 n36		0.56 0.41
11 b58	Brline PIPE	40 1	120 0.1016	1(us.Tee-Br); 1.524	0.1 1.624	50.22 1.5	0.0137 0.02	0.01 0.01	n59 h18	 50.22	0.22 0.19
12 b42	Brline PIPE	40 1	120 0.1016	1(us.Tee-Br); 1.524	0.1 1.624	51.2 1.53	0.0142 0.02	0.01 0.01	n43 h2	 51.2	0.23 0.2
13 b63	Brline PIPE	40 1	120 0.1016	1(us.Tee-Br); 1.524	0.1 1.624	51.21 1.53	0.0142 0.02	0.01 0.01	n63 h23	 51.21	0.23 0.2
14 b72	Brline PIPE	40 1	120 0.1016	1(us.90); 0.61	0.1 0.71	51.75 1.55	0.0145 0.01	0.01 0.01	n95 h32	 51.75	0.22 0.2
14 b40	Brline PIPE	40 1.5	120 0.1016	1(us.Tee-Run);	2.7	51.75 0.66	0.0018 0.00	0.00 0.03	n39 n95		0.25 0.22
14 b39	Brline PIPE	40 1.5	120 0.1016	1(us.Tee-Run);	2.7	105.68 1.34	0.0068 0.02	0.01 0.03	n38 n39		0.3 0.25
14 b38	Brline PIPE	40 1.5	120 0.1016	1(us.Tee-Run);	2.7	164.51 2.09	0.0153 0.04	0.02 0.03	n37 n38		0.37 0.3

Pipe Data

Path # Pipe Ref.	Type Hgroup	Schedule Size	HWC Rough.	Fittings Eq.Len.	Length Total Len.	Flow Velocity	Fr.Resist. Loss Frict.	Vel.Pres. Loss Elev.	Start End	Start Disch. End Disch.	Start Tot.Pres. End Tot.Pres.
			 mm	 m	m m	l/min m/s	bar/m bar	bar bar		l/min l/min	bar bar
14 b37	Brline PIPE	40 1.5	120 0.1016	1(us.90); 1.219	0.712 1.931	230.02 2.92	0.0285 0.06	0.04 0.01	n91 n37		0.43 0.37
14 b36	Brline PIPE	40 1.5	120 0.1016	1(us.Tee-Br); 2.438	0.683 3.121	230.02 2.92	0.0285 0.09	0.04 0.07	n88 n91		0.59 0.43
15 b66	Brline PIPE	40 1	120 0.1016	1(us.Tee-Br); 1.524	0.1 1.624	52.15 1.56	0.0147 0.02	0.01 0.01	n67 h26	 52.15	0.24 0.2
16 b71	Brline PIPE	40 1	120 0.1016	1(us.Tee-Br); 1.524	0.1 1.624	53.94 1.61	0.0157 0.03	0.01 0.01	n39 h31	 53.94	0.25 0.22
17 b46	Brline PIPE	40 1	120 0.1016	1(us.Tee-Br); 1.524	0.1 1.624	54.09 1.62	0.0157 0.03	0.01 0.01	n46 h6	 54.09	0.26 0.22
18 b86	Brline PIPE	40 1	120 0.1016	1(us.90); 0.61	0.1 0.71	54.17 1.62	0.0158 0.01	0.01 0.01	n101 h36	 54.17	0.24 0.22
18 b84	Brline PIPE	40 1.5	120 0.1016	1(us.Tee-Run);	2.7	54.17 0.69	0.002 0.01	0.00 0.03	n94 n101		0.27 0.24
18 b82	Brline PIPE	40 1.5	120 0.1016	1(us.Tee-Run);	2.7	110.35 1.4	0.0073 0.02	0.01 0.03	n93 n94		0.32 0.27
18 b80	Brline PIPE	40 1.5	120 0.1016	1(us.Tee-Run);	2.7	171.39 2.17	0.0165 0.04	0.02 0.03	n92 n93		0.39 0.32
18 b79	Brline PIPE	40 1.5	120 0.1016	1(us.90); 1.219	0.712 1.931	239.2 3.04	0.0307 0.06	0.05 0.01	n97 n92		0.46 0.39
18 b78	Brline PIPE	40 1.5	120 0.1016	1(us.Tee-Br); 2.438	0.683 3.121	239.2 3.04	0.0307 0.1	0.05 0.07	n89 n97		0.62 0.46
19 b51	Brline PIPE	40 1	120 0.1016	1(us.Tee-Br); 1.524	0.1 1.624	54.27 1.62	0.0158 0.03	0.01 0.01	n50 h11	 54.27	0.26 0.22
20 b54	Brline PIPE	40 1	120 0.1016	1(us.Tee-Br); 1.524	0.1 1.624	54.54 1.63	0.016 0.03	0.01 0.01	n54 h14	 54.54	0.26 0.22
21 b59	Brline PIPE	40 1	120 0.1016	1(us.Tee-Br); 1.524	0.1 1.624	55.18 1.65	0.0163 0.03	0.01 0.01	n58 h19	 55.18	0.27 0.23
22 b43	Brline PIPE	40 1	120 0.1016	1(us.Tee-Br); 1.524	0.1 1.624	56.14 1.68	0.0169 0.03	0.01 0.01	n42 h3	 56.14	0.27 0.24
23 b62	Brline PIPE	40 1	120 0.1016	1(us.Tee-Br); 1.524	0.1 1.624	56.14 1.68	0.0169 0.03	0.01 0.01	n62 h22	 56.14	0.27 0.24
24 b85	Brline PIPE	40 1	120 0.1016	1(us.Tee-Br); 1.524	0.1 1.624	56.18 1.68	0.0169 0.03	0.01 0.01	n94 h35	 56.18	0.27 0.24
25 b95	Brline PIPE	40 1	120 0.1016	1(us.90); 0.61	0.1 0.71	56.21 1.68	0.0169 0.01	0.01 0.01	n40 h40	 56.21	0.26 0.24
25 b93	Brline PIPE	40 1.5	120 0.1016	1(us.Tee-Run);	2.7	56.21 0.71	0.0021 0.01	0.00 0.03	n100 n40		0.29 0.26
25 b91	Brline PIPE	40 1.5	120 0.1016	1(us.Tee-Run);	2.7	114.3 1.45	0.0078 0.02	0.01 0.03	n99 n100		0.34 0.29
25 b89	Brline PIPE	40 1.5	120 0.1016	1(us.Tee-Run);	2.7	177.24 2.25	0.0176 0.05	0.03 0.03	n98 n99		0.42 0.34
25 b88	Brline PIPE	40 1.5	120 0.1016	1(us.90); 1.219	0.712 1.931	247 3.13	0.0325 0.06	0.05 0.01	n22 n98		0.49 0.42
25 b87	Brline PIPE	40 1.5	120 0.1016	1(us.Tee-Br); 2.438	0.683 3.121	247 3.13	0.0325 0.1	0.05 0.07	n96 n22		0.66 0.49
26 b67	Brline PIPE	40 1	120 0.1016	1(us.Tee-Br); 1.524	0.1 1.624	57.07 1.71	0.0174 0.03	0.01 0.01	n66 h27	 57.07	0.28 0.24
27 b94	Brline PIPE	40 1	120 0.1016	1(us.Tee-Br); 1.524	0.1 1.624	58.09 1.74	0.018 0.03	0.02 0.01	n100 h39	 58.09	0.29 0.25

Pipe Data

Path # Pipe Ref.	Type Hgroup	Schedule Size	HWC Rough.	Fittings Eq.Len.	Length Total Len.	Flow Velocity	Fr.Resist. Loss Frict.	Vel.Pres. Loss Elev.	Start End	Start Disch. End Disch.	Start Tot.Pres. End Tot.Pres.
			mm	m	m m	l/min m/s	bar/m bar	bar bar		l/min l/min	bar bar
28 b70	Brline PIPE	40 1	120 0.1016	1(us.Tee-Br); 1.524	0.1 1.624	58.83 1.76	0.0184 0.03	0.02 0.01	n38 h30	 58.83	0.3 0.26
29 b45	Brline PIPE	40 1	120 0.1016	1(us.Tee-Br); 1.524	0.1 1.624	60.63 1.81	0.0194 0.03	0.02 0.01	n45 h5	 60.63	0.32 0.28
30 b52	Brline PIPE	40 1	120 0.1016	1(us.Tee-Br); 1.524	0.1 1.624	60.82 1.82	0.0196 0.03	0.02 0.01	n49 h12	 60.82	0.32 0.28
31 b83	Brline PIPE	40 1	120 0.1016	1(us.Tee-Br); 1.524	0.1 1.624	61.05 1.82	0.0197 0.03	0.02 0.01	n93 h34	 61.05	0.32 0.28
32 b53	Brline PIPE	40 1	120 0.1016	1(us.Tee-Br); 1.524	0.1 1.624	61.09 1.83	0.0197 0.03	0.02 0.01	n53 h13	 61.09	0.32 0.28
33 b60	Brline PIPE	40 1	120 0.1016	1(us.Tee-Br); 1.524	0.1 1.624	61.75 1.85	0.0201 0.03	0.02 0.01	n57 h20	 61.75	0.33 0.29
34 b44	Brline PIPE	40 1	120 0.1016	1(us.Tee-Br); 1.524	0.1 1.624	62.74 1.88	0.0207 0.03	0.02 0.01	n41 h4	 62.74	0.34 0.3
35 b61	Brline PIPE	40 1	120 0.1016	1(us.Tee-Br); 1.524	0.1 1.624	62.74 1.88	0.0207 0.03	0.02 0.01	n61 h21	 62.74	0.34 0.3
36 b92	Brline PIPE	40 1	120 0.1016	1(us.Tee-Br); 1.524	0.1 1.624	62.94 1.88	0.0208 0.03	0.02 0.01	n99 h38	 62.94	0.34 0.3
37 b68	Brline PIPE	40 1	120 0.1016	1(us.Tee-Br); 1.524	0.1 1.624	63.69 1.9	0.0213 0.03	0.02 0.01	n65 h28	 63.69	0.35 0.31
38 b69	Brline PIPE	40 1	120 0.1016	1(us.Tee-Br); 1.524	0.1 1.624	65.51 1.96	0.0224 0.04	0.02 0.01	n37 h29	 65.51	0.37 0.32
39 b81	Brline PIPE	40 1	120 0.1016	1(us.Tee-Br); 1.524	0.1 1.624	67.8 2.03	0.0239 0.04	0.02 0.01	n92 h33	 67.8	0.39 0.35
40 b90	Brline PIPE	40 1	120 0.1016	1(us.Tee-Br); 1.524	0.1 1.624	69.76 2.09	0.0252 0.04	0.02 0.01	n98 h37	 69.76	0.42 0.37

PIPE INFORMATION

Node 1 Node 2	Elev 1 Elev 2	K-Factor 1 K-Factor 2	Flow added(q)* Total flow (Q)	Nominal ID Actual ID	Fittings quantity x (name) = length	L F T	C Factor Pf per m	total (Pt) elev (Pe) frict (Pf)	NOTES
	(m)	(lpm/bar½)	(l/min)	(mm)	(m)	(m)	(bar)	(bar)	
h8 n48	13.069 12.969	115.3	46.51 46.51	1 26.6	1x(us.90)=0.61	0.1 0.61 0.71	120 0.0119	0.16 0.01 0.01	
n48 n47	12.969 12.687		0 46.51	1.5 40.9		2.7 0 2.7	120 0.0015	0.18 0.03 0.00	
n47 n46	12.687 12.405		49.11 95.62	1.5 40.9		2.7 0 2.7	120 0.0056	0.21 0.03 0.02	
n46 n45	12.405 12.122		54.09 149.71	1.5 40.9		2.7 0 2.7	120 0.0129	0.26 0.03 0.03	
n45 n28	12.122 12.048		60.63 210.34	1.5 40.9	1x(us.90)=1.219	0.712 1.219 1.931	120 0.0242	0.32 0.01 0.05	
n28 n75	12.048 11.365		0 210.34	1.5 40.9	1x(us.Tee-Br)=2.438	0.683 2.438 3.121	120 0.0242	0.37 0.07 0.08	
n75 n74	11.365 11.365		218.87 429.2	4 102.3		2.6 0 2.6	120 0.001	0.51 0 0.00	
n74 n73	11.365 11.365		211.1 640.3	4 102.3		1.8 0 1.8	120 0.0022	0.52 0 0.00	
n73 n72	11.365 11.365		212.2 852.5	4 102.3		2.6 0 2.6	120 0.0037	0.52 0 0.01	
n72 n86	11.365 11.365		214.88 1067.38	4 102.3		2.6 0 2.6	120 0.0056	0.53 0 0.01	
n86 n71	11.365 11.365		218.88 1286.26	4 102.3		1.8 0 1.8	120 0.008	0.55 0 0.01	
n71 n88	11.365 11.365		222.72 1508.98	4 102.3		2.6 0 2.6	120 0.0107	0.56 0 0.03	
n88 n89	11.365 11.365		230.02 1739	4 102.3		2.6 0 2.6	120 0.0139	0.59 0 0.04	
n89 n96	11.365 11.365		239.2 1978.2	4 102.3		1.8 0 1.8	120 0.0177	0.62 0 0.03	
n96 n69	11.365 11.365		247 2225.19	4 102.3		60.85 0 60.85	120 0.022	0.66 0 1.34	
n69 n13	11.365 11.365		0 2225.19	6 154.1	1x(us.90)=4.267	21.788 4.267 26.055	120 0.003	1.99 0 0.08	
n13 n10	11.365 11.365		0 2225.19	6 154.1	1x(us.90)=4.267	20.963 4.267 25.23	120 0.003	2.07 0 0.08	
n10 n12	11.365 1.5		0 2225.19	6 154.1		9.865 0 9.865	120 0.003	2.15 0.97 0.03	

PIPE INFORMATION

Node 1 Node 2	Elev 1 Elev 2	K-Factor 1 K-Factor 2	Flow added(q)* Total flow (Q)	Nominal ID Actual ID	Fittings quantity x (name) = length	L F T	C Factor Pf per m	total (Pt) elev (Pe) frict (Pf)	NOTES
	(m)	(lpm/bar½)	(l/min)	(mm)	(m)	(m)	(bar)	(bar)	
n12 n11	1.5 0.5		0 2225.19	6 0		1 0 1	 0.0197	3.14 0.1 0.02	AV-1 Check ***
n11 n9	0.5 -0.5		0 2225.19	6 154.1	1x(us.90)=4.267	1 4.267 5.267	120 0.003	3.26 0.1 0.02	
n9 n8	-0.5 -0.5		0 2225.19	6 154.1	1x(us.90)=4.267	0.446 4.267 4.713	120 0.003	3.37 0 0.01	
n8 n7	-0.5 -0.5		0 2225.19	6 154.1	1x(us.90)=4.267	5.821 4.267 10.088	120 0.003	3.39 0 0.03	
n7 n6	-0.5 -0.5		0 2225.19	6 154.1	1x(us.90)=4.267	5.473 4.267 9.74	120 0.003	3.42 0 0.03	
n6 n5	-0.5 -0.5		0 2225.19	6 154.1	1x(us.90)=4.267	7.635 4.267 11.902	120 0.003	3.45 0 0.04	
n5 n4	-0.5 3		0 2225.19	6 154.1	1x(us.90)=4.267	3.5 4.267 7.767	120 0.003	3.48 -0.34 0.02	
n4 n3	3 3		0 2225.19	6 154.1	1x(us.90)=4.267	5.29 4.267 9.557	120 0.003	3.16 0 0.03	
n3 n2	3 3		0 2225.19	6 154.1	1x(us.90)=4.267	2.687 4.267 6.954	120 0.003	3.19 0 0.02	
n2 src1	3 0		0 2225.19	6 154.1		3 0 3	120 0.003	3.21 0.29 0.01	
h9 n52	13.069 12.969	115.3	46.71 46.71	1 26.6	1x(us.90)=0.61	0.1 0.61 0.71	120 0.012	0.16 0.01 0.01	
n52 n51	12.969 12.687		0 46.71	1.5 40.9		2.7 0 2.7	120 0.0015	0.18 0.03 0.00	
n51 n50	12.687 12.405		49.3 96.01	1.5 40.9		2.7 0 2.7	120 0.0057	0.21 0.03 0.02	
n50 n49	12.405 12.122		54.27 150.28	1.5 40.9		2.7 0 2.7	120 0.013	0.26 0.03 0.04	
n49 n30	12.122 12.048		60.82 211.1	1.5 40.9	1x(us.90)=1.219	0.712 1.219 1.931	120 0.0243	0.32 0.01 0.05	
n30 n74	12.048 11.365		0 211.1	1.5 40.9	1x(us.Tee-Br)=2.438	0.683 2.438 3.121	120 0.0243	0.37 0.07 0.08	
h16 n56	13.069 12.969	115.3	47.01 47.01	1 26.6	1x(us.90)=0.61	0.1 0.61 0.71	120 0.0121	0.17 0.01 0.01	
n56 n55	12.969 12.687		0 47.01	1.5 40.9		2.7 0 2.7	120 0.0015	0.18 0.03 0.00	

PIPE INFORMATION

Node 1 Node 2	Elev 1 Elev 2	K-Factor 1 K-Factor 2	Flow added(q)* Total flow (Q)	Nominal ID Actual ID	Fittings quantity x (name) = length	L F T	C Factor Pf per m	total (Pt) elev (Pe) frict (Pf)	NOTES
	(m)	(lpm/bar½)	(l/min)	(mm)	(m)	(m)	(bar)	(bar)	
n55 n54	12.687 12.405		49.57 96.57	1.5 40.9		2.7 0 2.7	120 0.0057	0.22 0.03 0.02	
n54 n53	12.405 12.122		54.54 151.11	1.5 40.9		2.7 0 2.7	120 0.0131	0.26 0.03 0.04	
n53 n32	12.122 12.048		61.09 212.2	1.5 40.9	1x(us.90)=1.219	0.712 1.219 1.931	120 0.0246	0.32 0.01 0.05	
n32 n73	12.048 11.365		0 212.2	1.5 40.9	1x(us.Tee-Br)=2.438	0.683 2.438 3.121	120 0.0246	0.38 0.07 0.08	
h17 n60	13.069 12.969	115.3	47.72 47.72	1 26.6	1x(us.90)=0.61	0.1 0.61 0.71	120 0.0125	0.17 0.01 0.01	
n60 n59	12.969 12.687		0 47.72	1.5 40.9		2.7 0 2.7	120 0.0015	0.19 0.03 0.00	
n59 n58	12.687 12.405		50.22 97.95	1.5 40.9		2.7 0 2.7	120 0.0059	0.22 0.03 0.02	
n58 n57	12.405 12.122		55.18 153.13	1.5 40.9		2.7 0 2.7	120 0.0134	0.27 0.03 0.04	
n57 n34	12.122 12.048		61.75 214.88	1.5 40.9	1x(us.90)=1.219	0.712 1.219 1.931	120 0.0251	0.33 0.01 0.05	
n34 n72	12.048 11.365		0 214.88	1.5 40.9	1x(us.Tee-Br)=2.438	0.683 2.438 3.121	120 0.0251	0.38 0.07 0.08	
h1 n44	13.069 12.969	115.3	48.79 48.79	1 26.6	1x(us.90)=0.61	0.1 0.61 0.71	120 0.013	0.18 0.01 0.01	
n44 n43	12.969 12.687		0 48.79	1.5 40.9		2.7 0 2.7	120 0.0016	0.2 0.03 0.00	
n43 n42	12.687 12.405		51.2 99.99	1.5 40.9		2.7 0 2.7	120 0.0061	0.23 0.03 0.02	
n42 n41	12.405 12.122		56.14 156.13	1.5 40.9		2.7 0 2.7	120 0.0139	0.27 0.03 0.04	
n41 n26	12.122 12.048		62.74 218.87	1.5 40.9	1x(us.90)=1.219	0.712 1.219 1.931	120 0.026	0.34 0.01 0.05	
n26 n76	12.048 11.365		0 218.87	1.5 40.9	1x(us.90)=1.219	0.683 1.219 1.902	120 0.026	0.4 0.07 0.05	
n76 n75	11.365 11.365		0 218.87	4 102.3		2.6 0 2.6	120 0.0003	0.51 0 0	
h24 n64	13.069 12.969	115.3	48.79 48.79	1 26.6	1x(us.90)=0.61	0.1 0.61 0.71	120 0.013	0.18 0.01 0.01	

PIPE INFORMATION

Node 1 Node 2	Elev 1 Elev 2	K-Factor 1 K-Factor 2	Flow added(q)* Total flow (Q)	Nominal ID Actual ID	Fittings quantity x (name) = length	L F T	C Factor Pf per m	total (Pt) elev (Pe) frict (Pf)	NOTES
	(m)	(lpm/bar½)	(l/min)	(mm)	(m)	(m)	(bar)	(bar)	
n64 n63	12.969 12.687		0 48.79	1.5 40.9		2.7 0 2.7	120 0.0016	0.2 0.03 0.00	
n63 n62	12.687 12.405		51.21 100	1.5 40.9		2.7 0 2.7	120 0.0061	0.23 0.03 0.02	
n62 n61	12.405 12.122		56.14 156.14	1.5 40.9		2.7 0 2.7	120 0.0139	0.27 0.03 0.04	
n61 n87	12.122 12.048		62.74 218.88	1.5 40.9	1x(us.90)=1.219	0.712 1.219 1.931	120 0.026	0.34 0.01 0.05	
n87 n86	12.048 11.365		0 218.88	1.5 40.9	1x(us.Tee-Br)=2.438	0.683 2.438 3.121	120 0.026	0.4 0.07 0.08	
h7 n47	12.787 12.687	115.3	49.11 49.11	1 26.6	1x(us.Tee-Br)=1.524	0.1 1.524 1.624	120 0.0132	0.18 0.01 0.02	
h10 n51	12.787 12.687	115.3	49.3 49.3	1 26.6	1x(us.Tee-Br)=1.524	0.1 1.524 1.624	120 0.0133	0.18 0.01 0.02	
h15 n55	12.787 12.687	115.3	49.57 49.57	1 26.6	1x(us.Tee-Br)=1.524	0.1 1.524 1.624	120 0.0134	0.18 0.01 0.02	
h25 n68	13.069 12.969	115.3	49.81 49.81	1 26.6	1x(us.90)=0.61	0.1 0.61 0.71	120 0.0135	0.19 0.01 0.01	
n68 n67	12.969 12.687		0 49.81	1.5 40.9		2.7 0 2.7	120 0.0017	0.21 0.03 0.00	
n67 n66	12.687 12.405		52.15 101.96	1.5 40.9		2.7 0 2.7	120 0.0063	0.24 0.03 0.02	
n66 n65	12.405 12.122		57.07 159.03	1.5 40.9		2.7 0 2.7	120 0.0144	0.28 0.03 0.04	
n65 n36	12.122 12.048		63.69 222.72	1.5 40.9	1x(us.90)=1.219	0.712 1.219 1.931	120 0.0269	0.35 0.01 0.05	
n36 n71	12.048 11.365		0 222.72	1.5 40.9	1x(us.Tee-Br)=2.438	0.683 2.438 3.121	120 0.0269	0.41 0.07 0.08	
h18 n59	12.787 12.687	115.3	50.22 50.22	1 26.6	1x(us.Tee-Br)=1.524	0.1 1.524 1.624	120 0.0137	0.19 0.01 0.02	
h2 n43	12.787 12.687	115.3	51.2 51.2	1 26.6	1x(us.Tee-Br)=1.524	0.1 1.524 1.624	120 0.0142	0.2 0.01 0.02	
h23 n63	12.787 12.687	115.3	51.21 51.21	1 26.6	1x(us.Tee-Br)=1.524	0.1 1.524 1.624	120 0.0142	0.2 0.01 0.02	
h32 n95	13.069 12.969	115.3	51.75 51.75	1 26.6	1x(us.90)=0.61	0.1 0.61 0.71	120 0.0145	0.2 0.01 0.01	

PIPE INFORMATION

Node 1 Node 2	Elev 1 Elev 2	K-Factor 1 K-Factor 2	Flow added(q)* Total flow (Q)	Nominal ID Actual ID	Fittings quantity x (name) = length	L F T	C Factor Pf per m	total (Pt) elev (Pe) frict (Pf)	NOTES
	(m)	(lpm/bar½)	(l/min)	(mm)	(m)	(m)	(bar)	(bar)	
n95 n39	12.969 12.687		0 51.75	1.5 40.9		2.7 0 2.7	120 0.0018	0.22 0.03 0.00	
n39 n38	12.687 12.405		53.94 105.68	1.5 40.9		2.7 0 2.7	120 0.0068	0.25 0.03 0.02	
n38 n37	12.405 12.122		58.83 164.51	1.5 40.9		2.7 0 2.7	120 0.0153	0.3 0.03 0.04	
n37 n91	12.122 12.048		65.51 230.02	1.5 40.9	1x(us.90)=1.219	0.712 1.219 1.931	120 0.0285	0.37 0.01 0.06	
n91 n88	12.048 11.365		0 230.02	1.5 40.9	1x(us.Tee-Br)=2.438	0.683 2.438 3.121	120 0.0285	0.43 0.07 0.09	
h26 n67	12.787 12.687	115.3	52.15 52.15	1 26.6	1x(us.Tee-Br)=1.524	0.1 1.524 1.624	120 0.0147	0.2 0.01 0.02	
h31 n39	12.787 12.687	115.3	53.94 53.94	1 26.6	1x(us.Tee-Br)=1.524	0.1 1.524 1.624	120 0.0157	0.22 0.01 0.03	
h6 n46	12.505 12.405	115.3	54.09 54.09	1 26.6	1x(us.Tee-Br)=1.524	0.1 1.524 1.624	120 0.0157	0.22 0.01 0.03	
h36 n101	13.069 12.969	115.3	54.17 54.17	1 26.6	1x(us.90)=0.61	0.1 0.61 0.71	120 0.0158	0.22 0.01 0.01	
n101 n94	12.969 12.687		0 54.17	1.5 40.9		2.7 0 2.7	120 0.002	0.24 0.03 0.01	
n94 n93	12.687 12.405		56.18 110.35	1.5 40.9		2.7 0 2.7	120 0.0073	0.27 0.03 0.02	
n93 n92	12.405 12.122		61.05 171.39	1.5 40.9		2.7 0 2.7	120 0.0165	0.32 0.03 0.04	
n92 n97	12.122 12.048		67.8 239.2	1.5 40.9	1x(us.90)=1.219	0.712 1.219 1.931	120 0.0307	0.39 0.01 0.06	
n97 n89	12.048 11.365		0 239.2	1.5 40.9	1x(us.Tee-Br)=2.438	0.683 2.438 3.121	120 0.0307	0.46 0.07 0.1	
h11 n50	12.505 12.405	115.3	54.27 54.27	1 26.6	1x(us.Tee-Br)=1.524	0.1 1.524 1.624	120 0.0158	0.22 0.01 0.03	
h14 n54	12.505 12.405	115.3	54.54 54.54	1 26.6	1x(us.Tee-Br)=1.524	0.1 1.524 1.624	120 0.016	0.22 0.01 0.03	
h19 n58	12.505 12.405	115.3	55.18 55.18	1 26.6	1x(us.Tee-Br)=1.524	0.1 1.524 1.624	120 0.0163	0.23 0.01 0.03	
h3 n42	12.505 12.405	115.3	56.14 56.14	1 26.6	1x(us.Tee-Br)=1.524	0.1 1.524 1.624	120 0.0169	0.24 0.01 0.03	

PIPE INFORMATION

Node 1 Node 2	Elev 1 Elev 2	K-Factor 1 K-Factor 2	Flow added(q)* Total flow (Q)	Nominal ID Actual ID	Fittings quantity x (name) = length	L F T	C Factor Pf per m	total (Pt) elev (Pe) frict (Pf)	NOTES
	(m)	(lpm/bar½)	(l/min)	(mm)	(m)	(m)	(bar)	(bar)	
h22 n62	12.505 12.405	115.3	56.14 56.14	1 26.6	1x(us.Tee-Br)=1.524	0.1 1.524 1.624	120 0.0169	0.24 0.01 0.03	
h35 n94	12.787 12.687	115.3	56.18 56.18	1 26.6	1x(us.Tee-Br)=1.524	0.1 1.524 1.624	120 0.0169	0.24 0.01 0.03	
h40 n40	13.069 12.969	115.3	56.21 56.21	1 26.6	1x(us.90)=0.61	0.1 0.61 0.71	120 0.0169	0.24 0.01 0.01	
n40 n100	12.969 12.687		0 56.21	1.5 40.9		2.7 0 2.7	120 0.0021	0.26 0.03 0.01	
n100 n99	12.687 12.405		58.09 114.3	1.5 40.9		2.7 0 2.7	120 0.0078	0.29 0.03 0.02	
n99 n98	12.405 12.122		62.94 177.24	1.5 40.9		2.7 0 2.7	120 0.0176	0.34 0.03 0.05	
n98 n22	12.122 12.048		69.76 247	1.5 40.9	1x(us.90)=1.219	0.712 1.219 1.931	120 0.0325	0.42 0.01 0.06	
n22 n96	12.048 11.365		0 247	1.5 40.9	1x(us.Tee-Br)=2.438	0.683 2.438 3.121	120 0.0325	0.49 0.07 0.1	
h27 n66	12.505 12.405	115.3	57.07 57.07	1 26.6	1x(us.Tee-Br)=1.524	0.1 1.524 1.624	120 0.0174	0.24 0.01 0.03	
h39 n100	12.787 12.687	115.3	58.09 58.09	1 26.6	1x(us.Tee-Br)=1.524	0.1 1.524 1.624	120 0.018	0.25 0.01 0.03	
h30 n38	12.505 12.405	115.3	58.83 58.83	1 26.6	1x(us.Tee-Br)=1.524	0.1 1.524 1.624	120 0.0184	0.26 0.01 0.03	
h5 n45	12.222 12.122	115.3	60.63 60.63	1 26.6	1x(us.Tee-Br)=1.524	0.1 1.524 1.624	120 0.0194	0.28 0.01 0.03	
h12 n49	12.222 12.122	115.3	60.82 60.82	1 26.6	1x(us.Tee-Br)=1.524	0.1 1.524 1.624	120 0.0196	0.28 0.01 0.03	
h34 n93	12.505 12.405	115.3	61.05 61.05	1 26.6	1x(us.Tee-Br)=1.524	0.1 1.524 1.624	120 0.0197	0.28 0.01 0.03	
h13 n53	12.222 12.122	115.3	61.09 61.09	1 26.6	1x(us.Tee-Br)=1.524	0.1 1.524 1.624	120 0.0197	0.28 0.01 0.03	
h20 n57	12.222 12.122	115.3	61.75 61.75	1 26.6	1x(us.Tee-Br)=1.524	0.1 1.524 1.624	120 0.0201	0.29 0.01 0.03	
h4 n41	12.222 12.122	115.3	62.74 62.74	1 26.6	1x(us.Tee-Br)=1.524	0.1 1.524 1.624	120 0.0207	0.3 0.01 0.03	
h21 n61	12.222 12.122	115.3	62.74 62.74	1 26.6	1x(us.Tee-Br)=1.524	0.1 1.524 1.624	120 0.0207	0.3 0.01 0.03	

PIPE INFORMATION

Node 1 Node 2	Elev 1 Elev 2	K-Factor 1 K-Factor 2	Flow added(q)* Total flow (Q)	Nominal ID Actual ID	Fittings quantity x (name) = length	L F T	C Factor Pf per m	total (Pt) elev (Pe) frict (Pf)	NOTES
	(m)	(lpm/bar½)	(l/min)	(mm)	(m)	(m)	(bar)	(bar)	
h38 n99	12.505 12.405	115.3	62.94 62.94	1 26.6	1x(us.Tee-Br)=1.524	0.1 1.524 1.624	120 0.0208	0.3 0.01 0.03	
h28 n65	12.222 12.122	115.3	63.69 63.69	1 26.6	1x(us.Tee-Br)=1.524	0.1 1.524 1.624	120 0.0213	0.31 0.01 0.03	
h29 n37	12.222 12.122	115.3	65.51 65.51	1 26.6	1x(us.Tee-Br)=1.524	0.1 1.524 1.624	120 0.0224	0.32 0.01 0.04	
h33 n92	12.222 12.122	115.3	67.8 67.8	1 26.6	1x(us.Tee-Br)=1.524	0.1 1.524 1.624	120 0.0239	0.35 0.01 0.04	
h37 n98	12.222 12.122	115.3	69.76 69.76	1 26.6	1x(us.Tee-Br)=1.524	0.1 1.524 1.624	120 0.0252	0.37 0.01 0.04	

* Discharge shown for flowing nodes only